



COMSATS University Islamabad

Department of Computer Science

Course Syllabus

Course Information

Course Code: **CSC262**

Credit Hours: **3 (2,1)**

Lab Hours/Week: **3**

Course Title: **Artificial Intelligence**

Lecture Hours/Week: **2**

Pre-Requisites: **None**

Catalogue Description:

This course gives a broad overview of the fundamental theories and techniques of Artificial Intelligence. Topics include: Artificial Intelligence Foundation; Problem-Solving; Adversarial Search; Constraint Satisfaction Problems; Knowledge Representation & Reasoning; Uncertainty and Probabilistic Reasoning; AI Domains Overview; AI Ethics and Future Trends.

Text and Reference Books

Textbook:

1. Artificial Intelligence: A Modern Approach, Russell, S., and Norvig, P., Pearson, 2020.

Reference Book:

1. Artificial Intelligence Basics: A Non-Technical Introduction, Taulli, T., Apress, 2019.

Week wise Plan:

Lecture #	CDF Unit #	Topics Covered	Reading Material
1.	1	What Is AI?; The Foundations of Artificial Intelligence; The History of Artificial Intelligence; The State of the Art	AIMA: Ch1
2.	1	Applications of AI: (Medicine, education, military, agriculture) Risks and Benefits of AI	AIMA: Ch1
3.	1	Intelligent Agent: The Concept of Rationality; Agents & Environments: The Nature of Environments; Agent Environments (Observable Agents, Deterministic, Episodic Static, and Discrete).	AIMA: Ch2
4.	1	The Structure of Agents: Simple reflex agents, Goal-based Agents, Model-based Reflex Agents, Utility-based Agents, and Learning Agents.	AIMA: Ch2
5.	2	Problem-Solving Agents; Search Algorithms: Best-first search, Depth-limited & iterative Deepening Search, Bidirectional search; Comparing uninformed search algorithms	AIMA: Ch3
6.	2	Informed (Heuristic) Search Strategies; Greedy best-first search; A* search; Search contours; Satisficing search: Inadmissible heuristics and weighted A*;Memory-bounded search; Bidirectional heuristic search	AIMA: Ch3
7.	2	Heuristic Functions: The effect of heuristic accuracy on performance; Generating heuristics from relaxed problems; Generating heuristics from subproblems: Pattern databases ; Generating heuristics with landmarks.	AIMA: Ch3
8.	2	Search in Complex Environments: Local Search and	AIMA: Ch3

		Optimization Problems: Hill-climbing search; Simulated annealing; Local beam search; Evolutionary algorithms; Local Search in Continuous Spaces; Search with Nondeterministic Actions	
9.	3	Adversarial Search and Games: Game Theory; Two-player zero-sum games; Optimal Decisions in Games	AIMA: Ch6
10.	3	Heuristic Alpha--Beta Tree Search: Evaluation functions; Cutting off search; Forward pruning ; Search versus lookup;	AIMA: Ch6
11.	3	Monte Carlo Tree Search, Selection, Expansion, Simulation, and Back Propagation.	AIMA: Ch6
12.	3	Partially Observable Games: Kriegspiel: Partially observable chess; Card games; Limitations of Game Search Algorithms.	AIMA: Ch6
13.	4	Defining Constraint Satisfaction Problems: Map coloring; Job-shop scheduling; Variations on the CSP formalism	AIMA: Ch5
14.	4	Constraint Propagation: Inference in CSPs: Node Consistency, Arc Consistency, Path Consistency, K-Consistency, Global Constraints; and Sudoku	AIMA: Ch5
15.	4	Backtracking Search for CSPs: Variable and value ordering. Interleaving Search & Inference, and Constraint Learning; Local Search for CSP, Min-Conflicts Heuristics	AIMA: Ch5
16.	4	The Structure of Problems, Cutset Conditioning, Tree Decomposition, and Value Symmetry.	AIMA: Ch5
17.	5	Introduction to Knowledge, Knowledge Based Agent, Wumpus World, Propositional Theorem Proving, Effective Propositional Model Checking and Agents Based on Propositional Logic.	AIMA: Ch7
18.	Mid Term Exam		
19.			
20.	5	First Order Logic: Syntax & Semantics of First Order Logic, Using FOL, and Knowledge Engineering in First-Order Logic.	AIMA: Ch8
21.	5	Inference in First-Order-Logic: Propositional vs. First-Order Inference; Unification & First-Order Inference; Forward Chaining; Backward Chaining; and Resolution.	AIMA: Ch9
22.	5	Knowledge Representation: Ontological Engineering, Categories & Objects, Events, Mental Objects & Modal Logic; Reasoning Systems for Categories; and Reasoning with Default Information.	AIMA: Ch10
23.	6	Quantifying Uncertainty: Acting under Uncertainty; Inference Using Full Joint Distributions; Bayes' Rule and Its Use; Naive Bayes Models	AIMA: Ch12
24.	6	Probabilistic Reasoning, Semantics of Bayesian Networks, Exact Inference in Bayesian Networks, and	AIMA: Ch13

		Approximate Inference for Bayesian Networks.	
25.	6	Dynamic Bayesian Networks: Constructing DBNs; Exact inference in DBNs; Approximate inference in DBNs	AIMA: Ch14
26.	6	Time & Uncertainty, Inference in Temporal Models; and Hidden Markov Models.	AIMA: Ch14
27.	7	AI domains overview and Fundamentals of Machine Learning, concept of supervised, unsupervised and reinforcement learning	AIMA: Ch19
28.	7	Machine Learning: Forms of Learning; Model Selection and Optimization; The Theory of Learning; Linear Regression & Classification; Nonparametric Models; Developing Machine Learning Systems.	AIMA: Ch19
29.	7	Basic concepts of Natural Language Processing: Language Models, Grammar, Parsing, Natural Language Tasks	AIMA: Ch23
30.	7	Fundamentals of Computer Vision : Image Formation, Image Features, Image classification, Object Detection	AIMA: Ch25
31.	7	Basics of Robotics: Robot Hardware , Robotic Perception, Planning and control, problems solved by robots	AIMA: Ch26
32.	7	AI Ethics and Future: Limits of AI, Can Machines Think, Components & Architecture of AI	AIMA: Ch27-28

Final Term Exam

Graduate Attributes (GAs)

S.#	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify, formulate, research literature, and solve <i>complex</i> computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
10	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

Course Learning Outcomes (CLO)

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA
CLO's for Theory				
CLO-1	1	Describe the foundational concepts of Artificial Intelligence and the models of intelligent agents.	<i>Understanding</i>	2
CLO-2	2-4	Apply search strategies to solve problems in Artificial Intelligence.	<i>Applying</i>	2-4
CLO-3	5-6	Apply knowledge representation and reasoning	<i>Applying</i>	3

		under uncertainty to solve problems in Artificial Intelligence.		
CLO-4	7	Apply foundational techniques from key AI domains to solve real-world problems, considering ethical implications.	<i>Applying</i>	3
CLO for Lab				
CLO-5	2-6	Implement various AI techniques and algorithms to solve an AI problem.	<i>Applying</i>	2-5
CLO-6	1-6	Develop a medium size project using AI techniques in a team environment.	<i>Creating</i>	2-6

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5	CLO-6
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-	-
Assignments	-	Assignment 1&2 (CCP ¹)	Assignment 3 (CCP)	Assignment 4 (CCP)	-	-
Sessional 1 Exam	Sessional 1 Exam		-	-	Lab Sessional 1	-
Sessional 2 Exam	-	Sessional 2 Exam		-	Lab Sessional 2	-
Final Term Exam	Final Term Exam					Lab Project/ Lab Final Term Exam (CCP+CCA ²)

1. CCP : Complex Computing Problem

2: CCA: Complex Computing Activity

Note:

1. Instructor should design theory assignments labeled as Complex Computing Problems (CCP) in such a way that each problem clearly exhibits 2–3 CCP characteristics.
2. Instructor should design theory assignments labeled as Complex Computing Activity (CCA) in such a way that each problem clearly exhibits 2–3 CCA characteristics.

Policy & Procedures

- **Attendance Policy:** Every student must attend 80% of the lectures as well as laboratory in this course. The students falling short of required percentage of attendance of lectures/laboratory work, is not allowed to appear in the terminal examination.

- **Course Assessment:**

	Quizzes	Assignments	Mid Term Exam	Terminal Exam	Total
Theory (T)	15	10	25	50	100
Lab (L)	-	25	25	50	100
Final Marks (T+L)	(T/100) *75 + (L/100) *25				

- **Grading Policy:** The minimum passing marks for each course is 50% (In case of LAB; in addition to theory, student is also required to obtain 50% marks in the lab to pass the course). The correspondence between letter grades, credit points, and percentage marks at CUI is as follows:

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	F
Marks	>= 85	80 – 84	75 – 79	71 – 74	68 – 70	64 – 67	61 – 63	58 – 60	54 – 57	50-53	< 50
Cr. Point	3.67-4.00	3.34-3.66	3.01-3.33	2.67-3.00	2.34-2.66	2.01-2.33	1.67-2.00	1.31-1.66	1.01-1.30	0.10-1.00	0.00

- **Missing Exam:** No makeup exam will be given for final exam under any circumstance. When a student misses the mid-term exam for a legitimate reason (such as medical emergencies), his grade for this exam will be determined based on the Department policy. Further, the student must provide an official excuse within one week of the missed exam.
- **Academic Integrity:** All CUI policies regarding ethics apply to this course. The students are advised to discuss their grievances/problems with their counsellors or course instructor in a respectful manner.
- **Plagiarism Policy:** Plagiarism, copying and any other dishonest behaviour is prohibited by the rules and regulations of CUI. Violators will face serious consequences.